

4 hours on examples

Problem 1 (19ESLA2)

We claim the answer is all constant functions f , which clearly works.

FTSOC, if there are any other solutions, we'd have multiple values in the range of f .

Case 1: There's a value in the range of f that appears infinitely many times.

Let m be a value that appears infinitely many times.

If m is not the minimum of the range of f , we construct a surjective function g such that $m - 1$ is not in the range of $f + g$. Let $g(x) = -1$ for all x such that $f(x) < m$. For these values, $f(x) + g(x) < m - 1$. If there exists any x such that $f(x) > m$, we just let $g(x) = 1$ for all such x .

As before, let $a_1, a_2, a_3, \dots$ be an infinite sequence of terms where $f(a_i) = m$ and $|a_1| \leq |a_2| \leq \dots$. Assign the odd indexed terms to be $0, 1, 2, \dots$ and the even indexed terms to be $-2, -3, \dots$. Every integer appears in the range of g and $f(x) + g(x) \neq m - 1$ as desired.

The construction for if m is the minimum (and hence is not the maximum) is analogous except setting the odd indices as $0, -1, -2, \dots$ and the evens as $2, 3, \dots$ to exclude $m + 1$ from the range of $f + g$.

Case 2: No value in the range of f appears infinitely many times.

There can't be both a minimum and a maximum as that'd mean the range is finite. WLOG, let there not be a maximum. We may construct a surjective g that ensures -1 does not appear in the range of $f + g$. For all x such that $f(x) \leq 0$, we let $g(x) = -1434$. Let $0 < a_1 < a_2 < \dots$ be the positive integers in the range of f .

For all x such that $f(x) = a_{2k+1}$ for some non-negative k , we force $g(x) = k$. Clearly, $f(x) + g(x) > 0$. For all x such that $f(x) = a_{2k}$, we force $g(x) = -k$. $f(x) + g(x) > 0$ still because $f(x) = a_{2k} \geq 2k$.

Problem 3 (13APMO4)

Consider a directed graph where the elements of $A \cup B$ are nodes. Draw an edge from i to $i + a$ if $i + a \in A$ and an edge from i to $i - b$ if $i - b \in B$.

Claim: We cannot have both $i + a \in A; i - b \in B$

Proof: Note that the indegree of each vertex in $A \cap B$ is at most 1 since A and B are disjoint. The outdegree of each $A \cap B$ vertex is at least 1. Given that $A \cap B$ is finite, the sum of the indegrees must equate the sum of the outdegrees, all the indegrees and outdegrees must be exactly 1. $\square$

We show that in each connected component on the graph, the ratio of nodes belonging to A to the number of nodes belong to B is $\frac{b}{a}$. Since each outdegree and indegree is 1, each connected component on our graph must be a cycle.

Starting at a point i on the connected component, we follow the directed edges. Whenever we jump to a node with a higher value, the value increases by a and we land on a node in A . When we jump to a node of a lower value, the value decreases by b and we land on a node in B . Since we'll end up where we started after passing through every node, the ratio of the number of increases to the number of decreases is $\frac{b}{a}$, which is the ratio of the number of nodes in A and that of B .

5 clubs; 6.5 hours

Problem 4 (87IMO4)

Consider a graph G on the non-negative where $x \rightarrow f(x)$. No cycle lies on the graph since we know $f^{(2k)}(x) = x + 1987k$ is unbounded. Furthermore, it is impossible for any node to have indegree > 1 since $f(a) = f(b) \Rightarrow f(f(a)) = f(f(b)) \Leftrightarrow a = b$. Therefore, we have a collection of chains.

Note that if $f(b) \equiv b \pmod{1987}$, we'd eventually have a contradiction since clearly $f(b) \neq b$ for $f(f(b)) = b + 1987$, and if $f(b) = b + 1987k$, taking, $f^{(m+2)}(x) = f^{(m)}(x) + 1987$ will eventually lead to a collision where $f^{(2k)}(b) = b + 1987k$ if k is positive or $f^{(1-2k)}(b) = b$ if k is negative. In these cases, there'd be a cycle following the graph starting at b . (!)

We start with the set $S = \{1, \dots, 1987\}$ and perform the following action 993 times:

- Take the smallest value x still in S
- Let the chain on G starting at x be c_i . All sufficiently large numbers that are $\equiv f(x), x \pmod{1987}$ lie on this chain
- Remove x and $f(x) \pmod{1987}$

Eventually, there's only one d left in S . $f(d) \not\equiv d \pmod{1987}$. Taking sufficiently large $f^{(2k+1)}(d)$ we'll run into another chain. (!)

Problem 5 (17SLN3)

We claim the answer is all prime n .

We show that all prime n work. FTSOC, let it be true that for every index $1 \leq k \leq n$, one of the numbers $a_k, a_k + a_{k+1}, \dots, a_k + a_{k+1} + \dots + a_{k+n-1}$. Let the sum $s := a_1 + \dots + a_n$. We know that $s \not\equiv 0 \pmod n$. We construct the modular prefix sums array b where $b_1 = 0$ and $b_i \equiv a_1 + \dots + a_{i-1}$ for $1 \leq i \leq n$.

Note that, for $k + j < n$ and $k \geq 1$,

$$a_k + a_{k+1} + \dots + a_{k+j} \equiv b_{k+j+1} - b_k \pmod n$$

For $k + j \geq n$ and $k \geq 1$,

$$a_k + a_{k+1} + \dots + a_{k+j} \equiv s - (a_{k+j+1-n} + \dots + a_{k-1}) \equiv s - (b_k - b_{k+j+1-n}) \pmod n$$

We place a marker on b_1 to start. Whenever the marker is on b_i , we consider the numbers $a_i, a_i + a_{i+1}, \dots, a_i + \dots + a_{i+n-1}$. Since one of these is divisible by n , we select the smallest prefix sum that is divisible by n , $a_i + \dots + a_{i+m} \equiv 0 \pmod n$ and move the marker to b_{i+m+1} if $i + m + 1 \leq n$ or $b_{i+m+1-n}$ if otherwise.

If we moved the marker to b_{i+m+1} , then $a_i + \dots + a_{i+m} \equiv 0 \pmod n \Leftrightarrow b_{i+m+1} \equiv b_i \pmod n$.

If we moved the marker to $b_{i+m+1-n}$, then $a_i + \dots + a_{i+m} \equiv s - (b_i - b_{i+m+1-n}) \pmod n$ meaning $b_{i+m+1-n} \equiv b_i - s \pmod n$.

Finally, note that if $m \neq n$ because $a_i + \dots + a_{i+m} \equiv a_1 + \dots + a_n \not\equiv 0 \pmod n$ so the marker cannot have returned to where it started.

The marker started at $b_1 \equiv 0 \pmod n$ and must have jumped forward to some other $b_i \equiv 0 \pmod n$ for its first move. The marker will eventually revisit a previous vertex. The marker cannot revisit a previous vertex b_i without having changed its current position's value $\pmod n$ since it can only jump forward in that case.

However, as soon as the marker jumps back, since n is prime and hence ns is the smallest multiple of s that is a multiple of n , it must jump back n times to revisit a certain remainder modulo n . Therefore, the marker visits some $b_i \equiv k \pmod n$ for any $k \pmod n$ and has also visited two distinct elements of b that are $0 \pmod n$. In total, the marker visited at least $n + 1$ distinct b_i . (!)

We now show that all composite $n = xy$ for $x, y > 1$ doesn't work because we can take $(0, x, x, x, \dots, x)$.

10 clubs; 9 hours

Problem 8 (09SLN5)

Consider the directed graph G where edges go from each integer x to $T(x)$.

Claim: x satisfies $T^n(x) = x$ only if x is in a cycle of length $k \mid n$ on G .

Proof: If x is not in a cycle, taking $T(x)$ repeatedly will never allow us to return to x meaning $T^n(x) \neq x$. Since x is in a cycle of length k , we must have $k \mid n$ for us to return to x after k moves. $\square$

We let c_i to be the number of cycles with length i . Note that each c_i must be finite for $f(i)$ to be finite. Since i numbers lie on each cycle of length i , there are $i \cdot c_i$ numbers in total lying on a cycle with length i for each i .

We have,

$$P(n) = \sum_{k \mid n} k \cdot c_k$$

Note that, for any $n = pq$ for some primes p, q ,

$$\begin{aligned} P(n) &= pq \cdot c_{pq} + p \cdot c_p + q \cdot c_q + 1 \cdot c_1 \\ &= pq \cdot c_{pq} + P(p) + P(q) - P(1) \end{aligned}$$

Taking mod p , we see that $P(p) \equiv P(0) \pmod{p}$ and $P(n) \equiv P(0) \pmod{p}$. Therefore,

$$\begin{aligned} P(0) &\equiv 0 + P(0) + P(q) - P(1) \pmod{p} \\ &\Leftrightarrow P(1) \equiv P(q) \pmod{p} \end{aligned}$$

Taking p large, we must have $P(q) = P(1)$ for any prime q . Therefore, $Q(x) = P(x) - P(1)$ has infinitely many zero meaning P is constant.

Problem 11 (18TSTST2)

Define a non-directed graph G from the roadmap of Onewaynia as follows. Each vertex on G is a one-way road in Onewaynia (**not cities**). For a given road $A \rightarrow B$, we draw one edge between it and the other road $C \rightarrow B$ leading to B . We draw another edge between it and the other road $A \rightarrow D$ leading from A .

Note that the degree of each vertex on G is exactly two, so G consists of cycles. Moreover, the length of these cycles must be even since every pair of adjacent vertices on G represent two roads connecting to the same city, and whether the roads lead to or come from the same city must alternates.

Therefore, we may satisfy the problem conditions if and only if we independently close every other road along each cycle.

Since, for each even-lengthed cycle, we independently have two choices for which group of “every other road” to close and which group to open, the answer is a power of two.

16 clubs; 11 hours

Problem 12 (20USATST5)

We claim the answer is any n such that its prime factors are not the smallest k prime numbers.

Let $n = p_1^{k_1} p_2^{k_2} \dots p_l^{k_l}$ and let $a_k = P^{(k)}(0)$. Since $a \equiv b \pmod{z} \implies P(a) \equiv P(b) \pmod{z}$ on intpolys, the problem is essentially asking for a polynomial P and integer m such that $a_k \pmod{n}$ is completely periodic with period m . If we calculate the periods of $a_k \pmod{p_i^{k_i}}$ to be t_i , the cumulative period is simply $\text{lcm}(t_1, t_2, \dots, t_l)$.

We first prove sufficiency. Let $n = p_1^{k_1} p_2^{k_2} \dots p_l^{k_l}$ for some $p_1 < p_2 < \dots < p_l$. There exists some other $p_{l-1} < p' < p_l$. For each i , let an integer $r_i \equiv p_i^{-k_i} \pmod{p_i^{k_i}}$. This inverse exists since p_i, p_l are both primes. We also let $r' \equiv \frac{1}{(p'-1)!} \pmod{p_l^{k_l}}$. We may construct,

$$P(n) = \left(\prod r_i p_i^{k_i} \right) \left((x+1) + (n-p')r'(x)(x-1) \dots (x-(p'-2)) \right)$$

In particular, $P(n) \equiv 0 \pmod{p_i^{k_i}}$ for any $i < l$ meaning the periods $t_1 = \dots = t_{l-1} = 1$.

Taking modulo $p_l^{k_l}$,

$$\begin{aligned} P(n) &= \left(\prod r_i p_i^{k_i} \right) \left((x+1) + (n-p')r'(x)(x-1) \dots (x-(p'-2)) \right) \\ &\equiv (x+1) + (n-p')r'(x)(x-1) \dots (x-(p'-2)) \pmod{p_l^{k_l}} \end{aligned}$$

For $n \equiv 0, 1, \dots, p'-2 \pmod{p_l^{k_l}}$, we have $f(n) \equiv n+1 \pmod{p_l^{k_l}}$. For $n \equiv p'-1$,

$$P(p' - 1) \equiv (p') + (n - p')r'(p' - 1)! \equiv p' + (n - p') \equiv 0 \pmod{p_l^{k_l}}$$

Therefore, $P(0) \equiv 1; P(1) \equiv 2; \dots; P(p' - 2) \equiv p' - 1; P(p' - 1) \equiv 0 \pmod{p_l^{k_l}}$ and the period is $t_l = p'$. Overall, the period is p' meaning taking $m = p'$ suffices.

We now prove necessity. FTSOC, let $p_1, \dots, p_l$ be the first l primes. Consider the period modulo p_i , which must be an amount T_i that is less than or equal to p_i . However, if $T_i > 1$, then there is some earlier prime dividing T_i meaning $\gcd(T_i, n) \neq 1$. Since we must have $a_i \equiv 0 \pmod{p_i}$ to have $a_i \equiv 0 \pmod{p_i^{k_i}}$, we know $T_i \mid t_i$ meaning $\gcd(t_i, n) \neq 1$ as well. Overall, this would result in $\gcd(n, \text{lcm}(t_1, \dots, t_l)) > 1$ (!). Therefore, $T_i = 1$.

Knowing that $a_i \equiv 0 \pmod{p_i}$, we then consider the period modulo p_i^2 and induct to eventually see that t_i must be equal to 1 [detail intentionally omitted].

Overall, $\text{lcm}(t_1, \dots, t_l) = 1$ meaning $m = 1$ which is not allowed.

Problem 14 (EULER209)

Consider a graph G on F_2^4 where we draw an edge between (a, b, c, d) and $(b, c, d, a + bc)$ for every tuple (a, b, c, d) . The graph ends up being a collection of cycles, forming the following connected components:

1. $(0, 0, 0, 0) - (0, 0, 0, 0)$
2. $(0, 0, 0, 1) - (0, 0, 1, 0) - (0, 1, 0, 0) - (1, 0, 0, 0) - (0, 0, 0, 1)$
3. $(0, 1, 0, 1) - (1, 0, 1, 0) - (0, 1, 0, 1)$
4. $(0, 0, 1, 1) - (0, 1, 1, 0) - (1, 1, 0, 1) - (1, 0, 1, 1) - (0, 1, 1, 1) - (1, 1, 1, 1) - (1, 1, 1, 0) - (1, 1, 0, 0) - (1, 0, 0, 1) - (0, 0, 1, 1)$

The condition is satisfied if and only if no two adjacent elements on G both output 1 under application of f .

For the first connected component of only $(0, 0, 0, 0)$, we must have $f(0, 0, 0, 0) = 0$ for one way.

For the second connected component of four elements, we may set all four to zero, set three of them to zero, or set two non-adjacent to zero for 1, 4, 2 options respectively and 7 ways in total.

For the third connected component with two elements, we may set both to zero or one of them to zero for 3 in total.

For the final connected component with nine elements. We may casework to find $1 + 9 + 27 + 30 + 9 = 76$ ways in total.

Overall, the answer is $1 \cdot 7 \cdot 3 \cdot 76 = 1596$ since each connected component can be treated independently.

24 clubs; 14 hours

Problem 15 (20IMO4)

We claim the answer is $n^2 - n$. The construction will be provided later.

Consider a graph G_A where the stations are vertices, and each cable car of company A is a directed edge. Since the indegree and outdegree of every vertex is at least 1 and each cable car goes higher, we may follow the edges leading from and to each vertex to see that G_A consists of several directed chains (possibly with length 1).

Claim: The number of chains and cable cars always sums to n^2

Proof: Chains are trees, and each tree has one less edge than it has vertices. $\square$

There are $n^2 - k$ chains. Let the lengths of the chains be $x_1 + x_2 + \dots + x_{n^2-k} = n^2$ respectively for $x_1 \leq x_2 \leq \dots \leq x_{n^2-k}$.

Clearly, two stations are linked by company A if and only if they are part of the same chain since we can follow the chain up and stations that are parts of different chains are parts of different connected components.

Analogously, G_B the graph for company B also consists of chains. For the problem condition to be met, no two stations part of the same chain on G_B are part of the same chain on G_A . As a result, each of the x_{n^2-k} stations on the longest chain of G_A must belong to different chains on G_B meaning G_B has at least x_{n^2-k} chains. By Pigeonhole, $x_{n^2-k} \geq \frac{n^2}{n^2-k}$. Therefore, G_B has at most $n^2 - \frac{n^2}{n^2-k}$ cable cars.

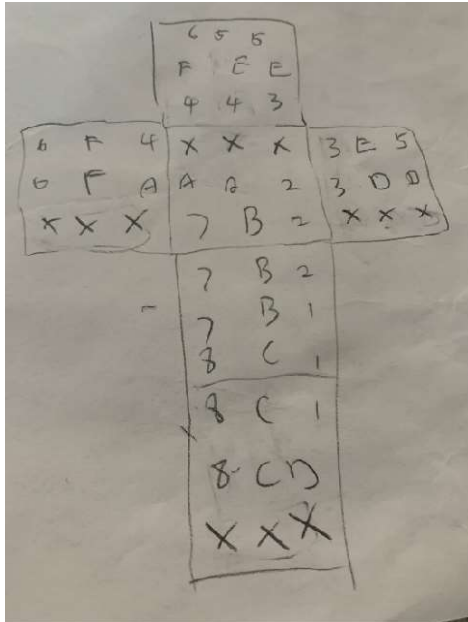
$$n^2 - \frac{n^2}{n^2-k} \leq k \Leftrightarrow n^2 \leq k + \frac{n^2}{n^2-k}$$

The RHS is strictly increasing, and we may check equality is achieved with $k = n^2 - n$. As a result, no greater k work.

One construction is to have G_A have chains $1 \rightarrow \dots \rightarrow n; n+1 \rightarrow \dots \rightarrow 2n; \dots$ and G_B have chains $1 \rightarrow n+1 \rightarrow 2n+1 \rightarrow \dots \rightarrow n^2-n+1; 2 \rightarrow n+2 \rightarrow \dots \rightarrow n^2-n+2; \dots$

Problem 18 (18MPO3)

The answer is 14, constructed as follows.



Each bent strip either (a) covers the center square of a face or (b) covers two of the three squares adjacent to a corner. There are 6 faces so at most 6 (a) types and 8 corners so at most 8 (b) types since we can't have two bent strips at the same corner for 14 in total. See other posts for construction.

32 clubs; 16.5 hours

Problem 19 (14IRNTST7)

Consider the directed graph G where the vertices are the vertices of T and each directed edge goes from the original position of each number on T to its destination. We show that graph G is just a cycle with length n . We induct on n .

Base case: $n = 1$. Nothing changes so cycle of 1 as desired.

Inductive step: After the first swap on edge $e: (x, y)$ in T , there is no more exchange between the two subgraphs A, B on the two sides of e since e was the only edge that connected A to B . Let $x \in A$ and $y \in B$. Note that $|A|, |B| \leq n - 1$. Therefore, by the inductive hypothesis, after this point, A, B will permute, and the permutation is a cycle with lengths $|A|, |B|$ respectively.

We define graphs G_A, G_B analogous to G but on A, B . To recreate G , we may merge graphs G_A, G_B then swap the destinations of the edges point to x and y . Following the cycle starting

at y , we will first visit every vertex on B . Whereas we would have returned to y on G_B , we'll be diverted to x and then visit every vertex on A before returning to y . We may check that this results in a cycle of length $|A| + |B| = n$ as desired. $\square$

Problem 20 (13ESLN6)

We claim the answer is all divisors of 2014.

Consider the directed graph G where each positive integer is a vertex and there is a directed edge from $x \rightarrow f(x)$. Since $f^{f(n)}(n)$ is finite and $f^{f^{f(n)}(n)}(n) = n$, we know each vertex lies on a directed cycle.

Claim: f is saturated if and only if each vertex on a cycle of length m is divisible by m .

Proof: To prove sufficiency, we have $f^{f^{f(n)}(n)}(n) = f^n(n) = n$.

For any n on the cycle, note that $f^{f^{f(n)}(n)}(n) = n$ means $f^{f(n)}(n)$ is a multiple of m . Since $f^{f(n)}(n)$ is also on the cycle, we know at least one vertex on the cycle is divisible by m . FTSOC, if not all of them are divisible by m , we'll then have a pair $(n, f(n))$ where $m \mid f(n)$ but not n . In that case,

$$f^{f^{f(n)}(n)}(n) = f^n(n) \neq n \quad \square$$

With the claim proven, we can prove that all divisors of 2014 work since the cycle involving any x dividing 2014 has length dividing x and hence dividing 2014.

To prove nothing else apart from the divisors of 2014 works, note that, for some d not dividing 2014, we may let,

$$f(d) = 2d; f(2d) = 3d; \dots; f(d^2 - d) = d^2; f(d^2) = d$$

For any other x not yet defined, we let $f(x) = x$. We check that this works since every $d, \dots, d^2$ lies on a cycle with length d while everything else lies on a cycle of length 1.

42 clubs; 19 hours

Problem 24 (12SLA6)

FTSOC, let there be a supremum M to the sequence.

Consider the sequence $a_n = f^{(n)}(1)$. For each a_n , there exists some $1 \leq k \leq M$ such that $a_{n+2k} = a_n + k$.

Claim 1: a_n cannot repeat terms.

Proof: Since $a_{n+1} = f(a_n)$, we know a_n will eventually become periodic if any term repeats. We cut out the non-repeating prefix and consider the maximum of the remaining repeating terms a_m . There is no term later in the sequence greater than a_m so $a_{m+2k} = a_m + k$ is impossible for finite k . $\square$

We define a directed graph G on the terms of a_n where each term a_n has exactly one directed edge coming out of it pointed to the earliest $a_{n+2k} = a_n + k$.

Claim 2: No two edges point to the same element of a .

Proof: Note that edges always span an even number of elements. FTSOC, if a_n is pointed to by both a_{n-2x} and a_{n-2y} for distinct x, y , we WLOG let $x < y$ and note that $a_{n-2x} = n - x$; $a_{n-2y} = n - y$ means $a_{n-2x} = a_{n-2y} + (x - y)$. Therefore, the edge from a_{n-2y} should have pointed to a_{n-2x} instead. $\square$

Note that there are clearly no cycles on G since each element points to a higher element in the sequence. Therefore, G is a collection of chains. We let the chains be $c_1, c_2, \dots$. No element a_i belongs to more than one chain so no integer belongs to more than one chain.

Let $c_{i,0} = a_{x_i}$ be the first element of c_i for some x_i and $c_{i,j}$ be the j -th element after the first in the i -th chain. Note that, by induction on j , $c_{i,j} = c_{i,0} + k \Leftrightarrow c_{i,j} = a_{x_i+2k}$.

$$\text{Plugging in } k = \frac{n}{2} - \frac{x_i}{2}, c_{i,j} = a_n \Leftrightarrow c_{i,j} = c_{i,0} + \frac{n}{2} - \frac{x_i}{2}. \quad (1)$$

Note that $1 \leq c_{i,j} - c_{i,j-1} \leq M$ meaning $c_{i,0} + j \leq c_{i,j} \leq c_{i,0} + Mj$. Therefore, out of the integers in $[1, N]$, at least $\frac{N-c_{i,0}}{M}$ belong to c_i and at most $N - c_{i,0} + 1$ belong to c_i . The proportion p_i of $[1, N]$ that belong to a certain chain c_i hence satisfies $\frac{1}{M} - \epsilon_2 \leq p_i \leq 1$ for arbitrarily small ϵ_2 as N gets large. As a result, since $\sum p_i \leq 1$ for no integer to be part of two chains, there are at most M chains.

Consider the range $\left[2 + \max_i(x_i - 2c_{i,0}), 2N + \min_i(x_i - 2c_{i,0})\right]$. We may use (1) and check that each a_k for $k \in \left[2 + \max_i(x_i - 2c_{i,0}), 2N + \min_i(x_i - 2c_{i,0})\right]$ must satisfy $1 \leq a_k \leq N$. Each element belonging to c_i here hence counted towards p_i . Since the size of $\left[2 + \max_i(x_i - 2c_{i,0}), 2N + \min_i(x_i - 2c_{i,0})\right]$ divided by N approaches 2 as N gets large due to $\max_i(x_i - 2c_{i,0}), \min_i(x_i - 2c_{i,0})$ being constants, if q_i is the proportion of a_k out of this range belonging to c_i , $q_i \leq \frac{p_i}{2} + \epsilon$ for arbitrarily small ϵ as N gets large. Each element of a_i belongs to a chain so $\sum q_i = 1$ must be true exactly. However, this implies,

$$\sum \frac{p_i}{2} + \epsilon \geq 1 \Rightarrow \sum \frac{p_i}{2} + \epsilon \geq \sum p_i$$

$$\Leftrightarrow \epsilon \geq \sum \frac{p_i}{2} \Rightarrow \epsilon \geq \frac{1}{2}M \cdot \left(\frac{1}{M} - \epsilon_2\right) \Leftrightarrow \epsilon + \frac{1}{2}M\epsilon_2 \geq \frac{1}{2}$$

This is clearly false since M is a constant and ϵ, ϵ_2 are arbitrarily small.

51 clubs; 21.5 hours